

BOTANY HUB MULTAI

B.Sc. I BOTANY — MINOR I

ELEMENTARY BOTANY

UNIT IV — Plant Physiology

DETAILED TEACHER-STYLE NOTES

Definitions • Every concept explained • Tables • Labelled study diagrams • Flowcharts • Examples • Comparisons • Exam points

Prepared in the same detailed format as Major Course II and Major Course III notes.

UNIT IV — PLANT PHYSIOLOGY

The supplied Minor I notes map this unit to imbibition, diffusion, osmosis, transport, transpiration, photosynthesis and respiration. ■filecite■turn7file0■L117-L137■

1. Imbibition

Definition — Imbibition: Absorption of water by hydrophilic colloidal substances, causing them to swell.

Example: dry seeds or dry wood absorb water and swell because cell-wall materials and other hydrophilic components attract water.

2. Diffusion

Definition — Diffusion: Net movement of particles from a region of higher concentration or chemical potential toward a region of lower concentration as a result of random molecular motion.

Factors affecting diffusion include concentration difference, temperature, surface area, distance and the nature of the medium.

3. Osmosis

Definition — Osmosis: Net movement of water across a selectively permeable membrane from a region of higher water potential toward lower water potential.

Process	What moves	Membrane needed?	Example
Imbibition	Water into hydrophilic material	No selectively permeable membrane required	Dry seed swelling
Diffusion	Particles/molecules	No special membrane required	Gas movement
Osmosis	Water	Yes, selectively permeable membrane	Cell water balance

4. Water absorption and transport

Root hairs provide a large contact area with soil water. Water enters root tissues and reaches xylem. In the shoot, transpiration contributes to the upward movement of water through the continuous water column.

Water pathway in a plant



5. Transpiration

Definition — Transpiration: Loss of water vapour from aerial parts of plants, mainly through stomata.

- Helps maintain the transpiration stream.
- Contributes to cooling.
- Can support mineral transport.
- Excessive water loss can create water stress; therefore stomatal regulation is important.

6. Photosynthesis

Definition — Photosynthesis: A process by which photosynthetic organisms use light energy to drive the formation of energy-rich organic compounds from carbon dioxide and water, with oxygen commonly released in oxygenic photosynthesis.

Conceptual photosynthesis flow



7. Respiration

Definition — Respiration: A set of metabolic processes in which organic molecules are broken down or transformed to generate ATP and other usable forms of energy for cellular work.

Feature	Photosynthesis	Respiration
Main role	Capture energy in organic compounds	Release usable energy from metabolic substrates
Major site	Chloroplast in plant cells	Mitochondria are major sites of aerobic stages
Gas relation	Net CO ₂ uptake and O ₂ release in typical oxygenic photosynthesis	Aerobic respiration uses O ₂ and releases CO ₂
Timing	Requires appropriate light reactions	Occurs continuously in living cells

Exam focus: Do not write 'photosynthesis is simply the reverse of respiration'. They are linked but involve different pathways, enzymes, compartments and energy transformations.